H1: Comprehensive Project, Problem Definition: Assessment Rubric

This rubric is shared with you so you know exactly what we're looking for. Your tutor uses the same rubric to assess your planning document.

How assessment works

H1 is a **gating contract** for H2. The outcome is **Signed off** (H2 may begin) or **Fix and Resubmit** (revise within the 2-attempt limit). Within-band grading (80, 85, 90+) happens later at H2 and H3. At H1 the question is simply: is this plan realistic enough that we can let you start building?

Your planning document is assessed on **five dimensions**. Each dimension is rated at one of three levels:

- **Fix and Resubmit**: revision needed before this dimension passes
- **Meets HD Standard**: this dimension passes
- **Exceeds**: this dimension is notably strong

To get signed off: You must reach "Meets HD Standard" on **all five dimensions**. If any dimension is "Fix and Resubmit," you'll be asked to revise.

Counts toward later marking: The number of "Exceeds" ratings on H1 is noted by your tutor and informs H2 expectations. A strong H1 may prompt your tutor to flag you for follow-up at H2 and, if the D4 + H2 evidence ultimately supports it, the **H+1 (High HD Recognition)** 90+ pathway. H1 by itself does *not* award H+1; H+1 is judged on D4 and H2 work (see `Tasks/TaskSpecs/H1plus_high_hd_recognition.md`).

The five dimensions

1. Problem definition (H1.1)

Is your problem specific, scoped, and worth solving with AI?

Fix and Resubmit	Meets HD Standard	Exceeds
Problem is vague ("use AI to help people") or framed as a learning exercise. No specific stakeholder. Success criteria are "make it work". Scope is unrealistic for 10–15 hours of H2 implementation.	Problem is specific. A named stakeholder makes a named decision. Scope fits 10–15 hours. Success criteria are observable. Why-AI argument addresses what's hard about the problem.	The problem statement reveals genuine understanding of where AI fits. You have explicitly narrowed an obviously-larger problem and can explain the cut.

What "specific" looks like: "A small-business owner uses the system to triage incoming customer emails into 'urgent / non-urgent / spam' so they spend their 30 minutes of email time on what matters" beats "AI for productivity".

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2. Risk register (H1.2)

Have you thought about what could go wrong, across data, technical, ethical, and evaluation dimensions?

Fix and Resubmit	Meets HD Standard	Exceeds
Fewer than 6 risks. Or missing one of the four required categories (data / technical / ethical / evaluation). Or risks are generic ("the model might be wrong"). Or mitigations are blank or "be careful".	At least 6 risks. All four categories present. Each risk is specific to *your* project. Mitigations are concrete (not generic advice).	You identify a non-obvious risk that signals real engagement with the problem. At least one mitigation has a measurable trigger ("if accuracy on segment X drops below Y, switch to fallback Z").

"Specific" test: Could your risk paragraph apply word-for-word to a completely different H2 project? If yes, it's too generic.

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3. Data sources (H1.3)

Have you identified concrete, accessible data, and thought honestly about its limits?

Fix and Resubmit	Meets HD Standard	Exceeds
Data source is unspecified, behind a paywall, requires consent you don't have, or is contradicted by your evaluation plan. No fallback. Biases section is "I don't think there are any".	Data source is named and accessible. Licence/consent appropriate. Biases or gaps named honestly. A workable fallback exists.	You have *looked* at the data and report a specific observation that shaped scope (e.g., "only 8% of rows have label X, so the evaluation will use stratified sampling"). The fallback is a coherent Plan B, not a downgrade.

Verification: If you claim to use a public dataset, your tutor will spot-check the link during the milestone discussion. Make sure the link works.

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4. Proposed solution (H1.4)

Is your technique realistic, justified, and grounded in FIT1061 concepts?

Fix and Resubmit	Meets HD Standard	Exceeds
Solution is a technique name with no design ("I'll use a neural network"). No alternative considered. Implementation steps are 1–2 bullets. Inputs/outputs unspecified.	Clear AI technique (or combination) from the unit. At least one alternative considered and rejected with a real reason. Inputs/outputs typed. Implementation steps decompose into 4–6 concrete chunks. Simplifying assumptions stated.	Technique justified against a *non-obvious* alternative (e.g., "weighted-sum over neural net for explainability, not just simplicity"). At least one implementation step is risk-driven: you'll probe the hardest part first.

Feasibility check: Mentally allocate 10–15 hours across your listed steps. If any single step would consume the whole budget, decompose it further.

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5. Evaluation methodology (H1.5)

Can your evaluation actually be run, and would it tell you when the system has failed?

Fix and Resubmit	Meets HD Standard	Exceeds
Evaluation is "see if it works". No baseline. No metric beyond accuracy on a single test. No failure-analysis approach. Success threshold absent or "as high as possible".	All five elements present: metric(s), baseline, test cases/split, failure analysis, success threshold. Threshold is *specific*: a number, a comparison, or an observable behaviour. Baseline is genuinely simpler than your solution.	The evaluation would reveal failure as informatively as success: you've thought about what would disconfirm your hypothesis. At least one metric is appropriate for the problem (not just accuracy) and connects to your H1.1 success criteria.

"Specific threshold" looks like: "Beats the majority-class baseline by ≥5 percentage points on macro-F1" beats "as high as possible".

Milestone discussion (W10)

After we sign off on the document, you'll have a ~15-minute milestone discussion with your tutor. They'll ask the six questions below (from the spec). Prepare answers: being able to defend the document fluently matters as much as the document itself.

1. Why is this problem suitable for an HD comprehensive project?
2. What is the simplest baseline, and why is your proposed solution more interesting?
3. What data assumption is most likely to fail?

4. Which risk in your register is most serious, and what will you do if it happens?
5. **How will your evaluation show that the system FAILED, not just that it succeeded?**
6. Who could be harmed or misled if this system were deployed and got things wrong?

Question 5 is the load-bearing one. Students who cannot answer it usually struggle with H2's evaluation dimension too. Take time on H1.5.

Quick self-check before submitting

Dimension	Minimum for "Meets HD Standard"
Problem definition	Specific stakeholder + specific decision + scoped + AI-justified
Risk register	6+ specific risks across all 4 categories, with concrete mitigations
Data sources	Named + accessible + biased-or-gapped honestly + has a fallback
Proposed solution	FIT1061 technique + alternative rejected + 4–6 steps + assumptions stated
Evaluation	Metric + baseline + test cases + failure analysis + specific threshold

If you can check all five, you're ready to submit. The milestone discussion will probe the document, so make sure you can defend every claim.

Before you upload, also work through the submission checklist at the bottom of `H1_planning_template.ipynb`:

- ☐ All five sections completed
- ☐ Risk register has at least six specific risks across all four categories
- ☐ AI use statement included (if you used AI to prepare H1)
- ☐ Document uploaded to OnTrack and marked **Ready for Feedback**
- ☐ Milestone discussion booked with your tutor

The 5-dimension self-check above tells you whether the content is HD-level. The submission checklist makes sure the administrative pieces are also in place.

What happens after sign-off

Once your H1 is signed off, you can start H2 implementation immediately; strong students use the 2-week overlap (W10–W12). H2 is graded against `Tasks/H2/student/H2_assessment_rubric.md`.

If you are aiming for **90+**, read `Tasks/TaskSpecs/H1plus_high_hd_recognition.md` (the H+1 High HD Recognition pathway). H+1 is *not* a new project: it's an evidence-based reflection on D4 + H2 work that bumps the HD mark to 90+ on the four markers in `unit_docs/assessment_faq.md`.

FIT1061 H1: Assessment Rubric (Student Version)